COGS 45: Computational Cognitive Science
Spring 2026

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Professor: Steven Frankland
Tuesday and Thursday 2:15-4:00.
Classroom: Irving 075
Office Hours: Friday 11 a.m.-12:00 p.m. in Raven 106, or by appointment.

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Synopsis

Human cognition is characterized by remarkable intellectual feats and equally frustrating failures. What principles determine what minds can and can't do? In this course, we treat human cognition as *computation* and construct artificial systems to model its strengths and weaknesses. We will explore how computational systems can learn (or fail to learn), how they can remember (and forget), and ways they can select appropriate (or inappropriate) actions.

We will cover basic principles and applications of neural networks and reinforcement learning, with readings from primary sources drawn broadly from psychology, artificial intelligence, and neuroscience. The first class each week (Tuesday) will be spent in lecture/discussion. The second class (Thursday) will be spent working directly with computational models in active learning exercises that focus on neural networks for hands-on exploration.

Learning Objectives

- Understanding prominent approaches to the computational modeling of human cognition.
- Broad familiarity with important contributions from computer science, artificial intelligence/machine learning, psychology, and neuroscience.
- Practical competence in building and evaluating neural networks, using Python.
- Deeper understanding of what's possible for each cognitive system, given objectives and constraints.

Course Prerequisites and Recommendations

COSC 1, or equivalent experience. (if uncertain, ask instructor)
Either COGS 1 or PSYC 28 are recommended, though not required.
Prior experience with calculus, probability, and linear algebra are helpful, but also not required.

Grading

30% Final Project (~10 pg paper).
50% Take-home Modeling Exercises
10% Midterm Quiz on Concepts
10% Course Attendance/Participation

Active-learning exercises

Modeling will be done in Python, using numpy and pytorch. Each week, we will work through a jupyter notebook that explores some of the topics in lecture. In addition to the parts we complete together, there will be a number of additional questions and modeling extensions. **Assignments are due within 2 weeks.**

The notebooks will be hosted using Google Colab. https://colab.research.google.com/.

Possible Reference Books for Labs **-** definitely not required reading, but could be helpful

Inside Deep Learning: Math, Algorithms, Models
PyTorch Pocket Reference: Building and Deploying Deep Learning Models

Final Project

~10 page paper (~2000 words). Choose one of two types of final project (other formats are possible; if you have an idea, talk to the instructor!).You can use the topics covered in this class as inspiration, but aren't constrained to – choose a topic you are most excited about. Due June 8th.

- *Option 1. Proposal* (*no additional modeling necessary).* Identify a question/phenomenon of interest to you regarding human cognition that is amenable to computational modeling. Conduct literature review on extant models. Hypothesize a particular computational mechanism/principle for the phenomenon, consider alternatives, and propose a plan for how to evaluate your proposed mechanism with modeling and, if necessary, empirical data. (helpful to use images to display predicted/possible outcomes),

- *Option 2. Modeling* (additional *modeling necessary).* (a) Take a question and/or model we have explored in class, and extend it in a new direction (e.g., new data, or new architecture, or new applications). Write-up your methods and results, with figures.

<u>Schedule</u>

Week 1.

Tuesday 03/31: Welcome! What is computational cognitive science?

Thursday 04/02: Lab. Introduction to Neural Networks with Colab

- Readings:

  Frank, M. C., & Goodman, N. D. (2025). Cognitive modeling using artificial intelligence. *Annual Review of Psychology*, *77*.

- Videos: 3blue1Brown - Learning using Gradient Descent
  https://www.youtube.com/watch?v=IHZwWFHWa-w&t=301s
  3blue1Brown - The Backpropagation algorithm
  https://www.youtube.com/watch?v=Ilg3gGewQ5U&t=401s

  Two optional background/context readings:
  - Griffiths, T. L., Chater, N., Kemp, C., Perfors, A., & Tenenbaum, J. B. (2010). Probabilistic models of cognition: Exploring representations and inductive biases. *Trends in cognitive sciences*, *14*(8), 357-364.
  - McClelland, J. L., Botvinick, M. M., Noelle, D. C., Plaut, D. C., Rogers, T. T., Seidenberg, M. S., & Smith, L. B. (2010). Letting structure emerge: connectionist and dynamical systems approaches to cognition. *Trends in cognitive sciences*, *14*(8), 348-356.

- **Homework Assignment: Learning with explicit teaching ("supervision").**

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Week 2.

Tuesday 04/07: Perceiving the world

Readings:

- Marr, D. (2010). *Vision: A computational investigation into the human representation and processing of visual information*. MIT press. **CHAPTER 1: The Philosophy and The Approach**
- DiCarlo, J. J., Zoccolan, D., & Rust, N. C. (2012). How does the brain solve visual object recognition?. *Neuron*, *73*(3), 415-434.

- Lindsay, G. W. (2021). Convolutional neural networks as a model of the visual system: Past, present, and future. *Journal of cognitive neuroscience*, *33*(10), 2017-2031.

Thursday 04/09- . Lab. Inside the black box: How do neural networks see the world?

Readings:

- Peterson, J. C., Abbott, J. T., & Griffiths, T. L. (2018). Evaluating (and improving) the correspondence between deep neural networks and human representations. *Cognitive science*, *42*(8), 2648-2669.
- Sucholutsky, I., Muttenthaler, L., Weller, A., Peng, A., Bobu, A., Kim, B., ... & Griffiths, T. L. (2023). Getting aligned on representational alignment. *arXiv preprint arXiv:2310.13018*. (**ONLY SECTIONS 1-3).**

Homework Assignment: More on Alignment with Deep Vision Models

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Week 3.

Tuesday, April 14th - Short & Long-Term Memory

Readings:

- Chater, N., Oaksford, M., Chater, N., & Oaksford, M. (1999). Ten years of the rational analysis of cognition. *Trends in cognitive sciences*, *3*(2), 57-65.
- Graves, A., Wayne, G., Reynolds, M., Harley, T., Danihelka, I., Grabska-Barwińska, A., ... & Hassabis, D. (2016). Hybrid computing using a neural network with dynamic external memory. *Nature*, *538*(7626), 471-476.
- Webb, T. W., Sinha, I., & Cohen, J. D. (2020). Emergent symbols through binding in external memory. *arXiv preprint arXiv:2012.14601*.

Thursday, April 16th:

- **In Class Lab: Rational Analysis, Catastrophic Forgetting, RNNs**
    - **RNN background**

McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Why there are complementary learning systems in the hippocampus and neocortex: insights from the successes and failures of connectionist models of learning and memory. *Psychological review*, *102*(3), 419. ((***long reading, ok to skim***).

- **Homework Assignment:** Shakespeare and RNNs

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Week 4.

Tuesday 04/21st - Memory 2: Memory for the future?

Readings:

- Schacter, D. L. (1999). The seven sins of memory: Insights from psychology and cognitive neuroscience. *American psychologist*, *54*(3), 182.
- Crane (2015) Computers and Thought, pp. 83-114.
- Graves et al. (2016 from last week.

Thursday 04/23rd: Lab. External Memory Systems & Efficient Generalization

- Webb et al. from last week.

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Week 5.

Tuesday April 28th: Varieties of "Attention": Human & Transformers

- Readings

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, *30*.

Lindsay, G. W. (2020). Attention in psychology, neuroscience, and machine learning. *Frontiers in computational neuroscience*, *14*, 29.

Thursday April, 30th: Lab. Transformers & "Induction Heads".

In-context Learning & Induction Heads

Week 6

Tuesday May 5th: Discussion - Learning and Using a Language

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021, March). On the dangers of stochastic parrots: Can language models be too big?🦜. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency* (pp. 610-623).

Piantadosi, S. T., & Hill, F. (2022). Meaning without reference in large language models. *arXiv preprint arXiv:2208.02957*.

Vong, W. K., Wang, W., Orhan, A. E., & Lake, B. M. (2024). Grounded language acquisition through the eyes and ears of a single child. *Science*, *383*(6682), 504-511.

Thursday May 7th: IN-CLASS QUIZ

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Week 7

Tuesday May 12th - How do we decide?

- Readings:

Gopnik, A. (2020). Childhood as a solution to explore–exploit tensions. *Philosophical Transactions of the Royal Society B*, *375*(1803), 20190502.

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., ... & Hassabis, D. (2015). Human-level control through deep reinforcement learning. nature, 518(7540), 529-533.

Thursday May 14th - Reinforcement Learning - learning with temporal differences

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Week 8.

Tuesday May 19th - Thinking with limited resources

Lieder, F., & Griffiths, T. L. (2020). Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and brain sciences*, *43*, e1.

Zheng, J., & Meister, M. (2025). The unbearable slowness of being: Why do we live at 10 bits/s?. *Neuron*, *113*(2), 192-204.

FAQ on LLM resource use

Thursday May 21st - “Vibe-Coding” a Lab on Resource Limits.

Jones, N. Vibe Coding in Science.

Diak, C.J., Nguyen, N., Tibbetts, J.R., Webb, T.W., & Frankland, S.M. (2026). Backward Digit Span Benchmarks Working Memory in LLMs. Proceedings of the Cognitive Science Society.

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Week 9.

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May 26th - Discussion

Chiang, T. ChatGPT is a blurry jpeg of the web. *New Yorker*. February 9th, 2023

Webb, T., Holyoak, K. J., & Lu, H. (2023). Emergent analogical reasoning in large language models. *Nature Human Behaviour*, *7*(9), 1526-1541.

Mitchell, M.. Artificial Intelligence Learns to Reason. https://www.science.org/doi/10.1126/science.adw5211

Griffiths, T. https://www.theguardian.com/books/2026/may/03/will-human-minds-still-be-special-in-an-age-of-ai

May 28th - Informal presentation/discussion of project ideas.

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Academic Honor Principle

The faculty, administration, and students of Dartmouth College acknowledge the responsibility to maintain and perpetuate the principle of academic honor, and recognize that any instance of academic dishonesty is considered a violation of the Academic Honor Principle.

Religious Observances

Dartmouth has a deep commitment to support students' religious observances and diverse faith practices. Some students may wish to take part in religious observances that occur during this academic term. If you have a religious observance that conflicts with your participation in the course, please meet with me as soon as possible—before the end of the second week of the term at the latest—to discuss appropriate course adjustments.

Student Accessibility and Accommodations

Students requesting disability-related accommodations and services for this course are required to register with Student Accessibility Services (SAS; Apply for Services webpage; student.accessibility.services@dartmouth.edu; 1-603-646-9900) and to request that an accommodation email be sent to me in advance of the need for an accommodation. Then, students should schedule a follow-up meeting with me to determine relevant details such as what role SAS or its Testing Center may play in accommodation implementation. This process works best for everyone when completed as early in the quarter as possible. If students have questions about whether they are eligible for accommodations or have concerns about the implementation of their accommodations, they should contact the SAS office. All inquiries and discussions will remain confidential.

Mental Health and Wellness

There are a number of resources available to you on campus to support your wellness, including: the Counseling Center which allows you to book triage appointments online, the Student Wellness Center which offers wellness check-ins, and your undergraduate dean. The student-led Dartmouth Student Mental Health Union and their peer support program may be helpful if you would like to speak to a trained fellow student support listener. If you need immediate assistance, please contact the counselor on-call at (603) 646-9442 at any time. Please make me aware of anything that will hinder your success in this course.

Title IX

At Dartmouth, we value integrity, responsibility, and respect for the rights and interests of others, all central to our Principles of Community. We are dedicated to establishing and maintaining a safe and inclusive campus where all community members have equal access to Dartmouth's educational and employment opportunities. We strive to promote an environment of sexual respect, safety, and well-being. Through the Sexual and Gender-Based Misconduct Policy (SMP), Dartmouth demonstrates that sex and gender-based discrimination, sex and gender-based harassment, sexual assault, dating violence, domestic violence, stalking, etc., are not tolerated in our community.

For more information regarding Title IX and to access helpful resources, visit Title IX's website (sexual-respect.dartmouth.edu). As a faculty member, I am required to share disclosures of sexual or gender-based misconduct with the Title IX office.

If you have any questions or want to explore support and assistance, please contact the Title IX office at 603-646-0922 or TitleIX@dartmouth.edu. Speaking to Title IX does not automatically initiate a college resolution. Instead, much of their work is around providing supportive measures to ensure you can continue to engage in Dartmouth's programs and activities.